

HOSPITAL DATA BASE



Made by :





THE ENTITIES

Of the hospital data base



Patient



Department



Manage



Docto



r
Nurse



Rooms



Employee



Appointmen



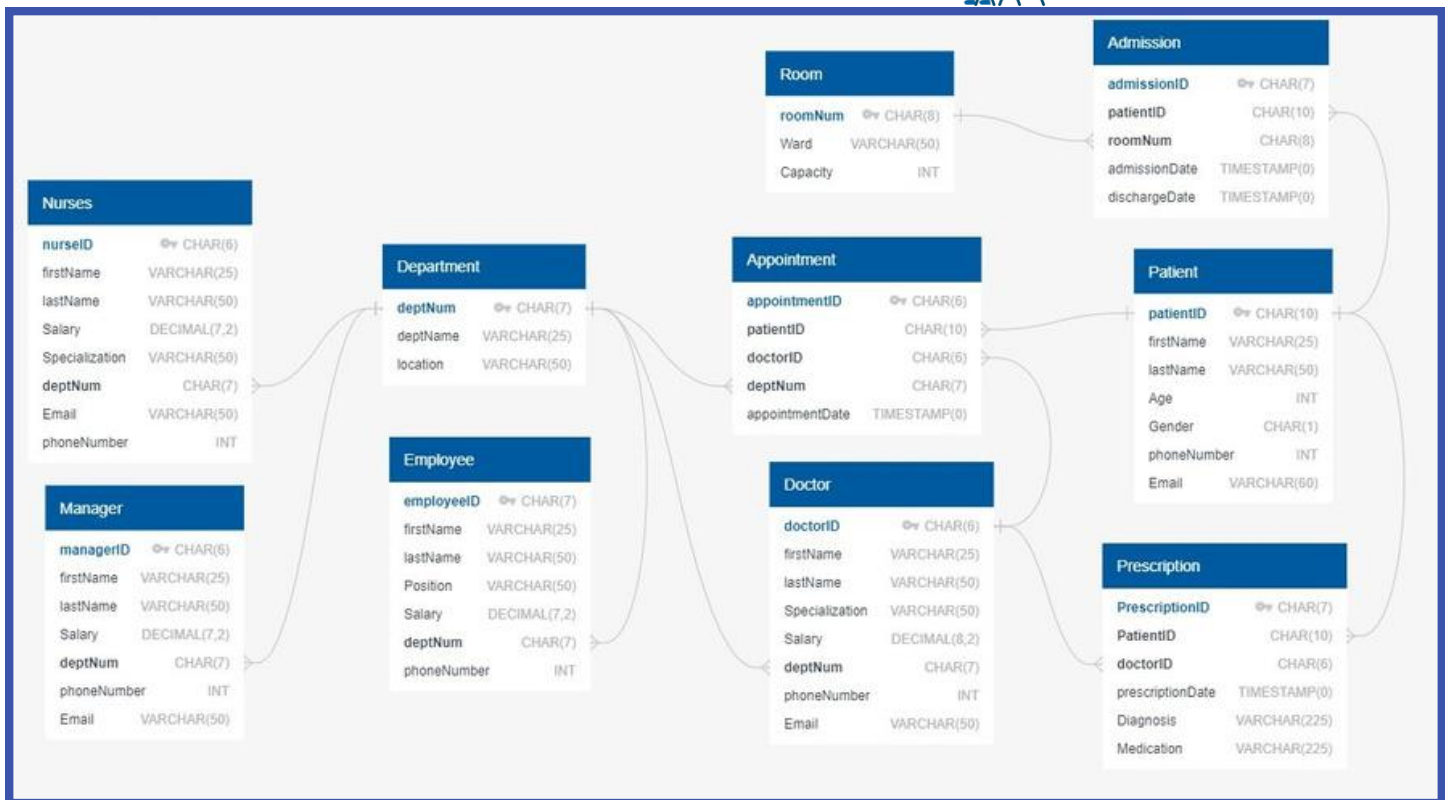
Admissio



n
Prescription



SCHEMA DIAGRAM





Data Definition Language(DDL)

```
CREATE TABLE Patient(  
patientID CHAR(10),  
firstName VARCHAR(25) NOT NULL, lastName  
VARCHAR(50) NOT NULL, Age INT NOT NULL  
CHECK(Age>=1), Gender CHAR(1) NOT NULL,  
phoneNumber INT NOT NULL,  
Email VARCHAR(60),  
CONSTRAINT pkPatient PRIMARY KEY  
(patientID)  
);
```



Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PATIENT	PATIENTID	CHAR	10	-	-	1	-	-	-
	FIRSTNAME	VARCHAR2	25	-	-	-	-	-	-
	LASTNAME	VARCHAR2	50	-	-	-	-	-	-
	AGE	NUMBER	22	-	0	-	-	-	-
	GENDER	CHAR	1	-	-	-	-	-	-
	PHONENUMBER	NUMBER	22	-	0	-	-	-	-
	EMAIL	VARCHAR2	60	-	-	-	✓	-	-
1 - 7									



Data Definition Language(DDL)

```
INSERT INTO Patient
VALUES ('P001', 'Abdullah', 'Mohammed', 35, 'M', 0535551234,
'abdullah.mohammed@email.com');
INSERT INTO Patient
VALUES ('P002', 'Fatima', 'Ali', 28, 'F', 0547895678, NULL);
INSERT INTO Patient
VALUES ('P003', 'Mohammed', 'Ahmed', 42, 'M', 0544349101, 'm_ahmed@gmail.com'); INSERT
INTO Patient
VALUES ('P004', 'Layla', 'Salman', 22, 'F', 0522121122, 'layla22@hotmail.com'); INSERT INTO
Patient
VALUES ('P005', 'Abdulaziz', 'Ali', 50, 'M', 0563353344, NULL);
INSERT INTO Patient
VALUES ('P006', 'Noura', 'Khaled', 31, 'F', 0513375566, 'nourakhaled@email.com'); INSERT INTO
Patient
VALUES ('P007', 'Sara', 'Abdullah', 27, 'F', 0553059900, 'sara_1997@gmail.com'); INSERT INTO
Patient
VALUES ('P008', 'Abdulrahman', 'Ahmed', 38, 'M', 0501851122, NULL);
INSERT INTO Patient
VALUES ('P009', 'Fatima', 'Mohammed', 29, 'F', 0597773344, 'fatima333@email.com'); INSERT
INTO Patient
VALUES ('P010', 'Khaled', 'Ali', 48, 'M', 0523325566, 'khaled_ali@email.com'); INSERT INTO
Patient
VALUES ('P011', 'Amal', 'Abdulrahman', 25, 'F', 0539997788, 'aaamal@email.com'); INSERT
INTO Patient
VALUES ('P012', 'Abdullah', 'Ali', 33, 'M', 0523219900, 'abdullah.ali@email.com'); INSERT INTO
Patient
VALUES ('P013', 'Nour', 'Ahmed', 39, 'F', 0510601122, NULL);
INSERT INTO Patient
VALUES ('P014', 'Mohammed', 'Abdullah', 26, 'M', 0579783344, NULL);
INSERT INTO Patient
VALUES ('P015', 'Lama', 'Ali', 36, 'F', 0586645566, 'lamaali@email.com');
```



Data Definition Language(DDL)



PATIENTID	FIRSTNAME	LASTNAME	AGE	GENDER	PHONENUMBER	EMAIL
P001	Abdullah	Mohammed	35	M	535551234	abdullah.mohammed@email.com
P002	Fatima	Ali	28	F	547895678	-
P003	Mohammed	Ahmed	42	M	544349101	m_ahmed@gmail.com
P004	Layla	Salman	22	F	522121122	layla22@hotmail.com
P005	Abdulaziz	Ali	50	M	563353344	-
P006	Noura	Khaled	31	F	513375566	nourakhaled@email.com
P007	Sara	Abdullah	27	F	553059900	sara_1997@gmail.com
P008	Abdulrahman	Ahmed	38	M	501851122	-
P009	Fatima	Mohammed	29	F	597773344	fatima333@email.com
P010	Khaled	Ali	48	M	523325566	khaled_ali@email.com
P011	Amal	Abdulrahman	25	F	539997788	aaamal@email.com
P012	Abdullah	Ali	33	M	523219900	abdullah.ali@email.com
P013	Nour	Ahmed	39	F	510601122	-
P014	Mohammed	Abdullah	26	M	579783344	-
P015	Lama	Ali	36	F	586645566	lamaali@email.com



Data Definition Language(DDL)

```
CREATE TABLE Department (  
  deptNum CHAR(7),  
  deptName Varchar(25) NOT NULL,  
  location VARCHAR(50),  
  CONSTRAINT pkDept PRIMARY KEY (deptNum)  
);
```

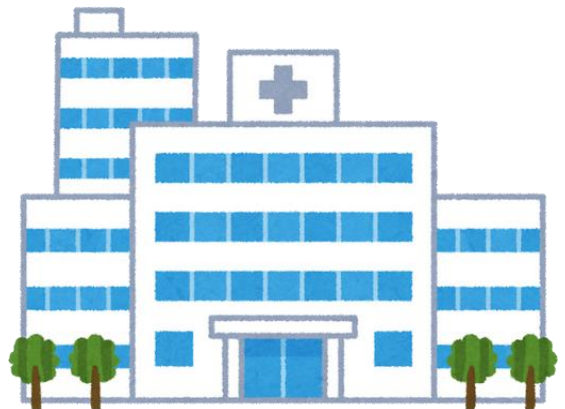


Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DEPARTMENT	DEPTNUM	CHAR	7	-	-	1	-	-	-
	DEPTNAME	VARCHAR2	25	-	-	-	-	-	-
	LOCATION	VARCHAR2	50	-	-	-	✓	-	-
									1 - 3



Data Definition Language(DDL)

```
insert into Department
values ('Dept001','Ophthalmology Department','Floor 2');
insert into Department
values ('Dept002','Orthopedics Department','Floor 2'); insert
into Department
values ('Dept003','Psychiatry Department','Floor 1'); insert into
Department
values ('Dept004','Cardiology Department','Floor 3'); insert into
Department
values ('Dept005','Dental Department ','Floor 5');
```

DEPTNUM	DEPTNAME	LOCATION
Dept001	Ophthalmology Department	Floor 2
Dept002	Orthopedics Department	Floor 2
Dept003	Psychiatry Department	Floor 1
Dept004	Cardiology Department	Floor 3
Dept005	Dental Department	Floor 5



Data Definition Language(DDL)

```
CREATE TABLE Manager (  
  managerID CHAR(6),  
  firstName VARCHAR(25) NOT NULL,  
  lastName VARCHAR(50) NOT NULL,  
  Salary DECIMAL(7,2) NOT NULL CHECK(Salary  
  between 6000 and 30000) ,  
  deptNum CHAR(7),  
  phoneNumber INT UNIQUE NOT NULL,  
  Email VARCHAR(50) ,  
  CONSTRAINT pkManager PRIMARY KEY  
  (managerID),  
  CONSTRAINT fkDeptMan FOREIGN KEY (deptNum)  
  REFERENCES Department (deptNum) );
```



Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
MANAGER	MANAGERID	CHAR	6	-	-	1	-	-	-
	FIRSTNAME	VARCHAR2	25	-	-	-	-	-	-
	LASTNAME	VARCHAR2	50	-	-	-	-	-	-
	SPECIALIZATION	VARCHAR2	50	-	-	-	-	-	-
	SALARY	NUMBER	-	8	2	-	-	-	-
	DEPTNUM	CHAR	7	-	-	-	✓	-	-
	PHONENUMBER	NUMBER	22	-	0	-	-	-	-
	EMAIL	VARCHAR2	50	-	-	-	✓	-	-
1 - 8									



Data Definition Language (DDL)

```
INSERT INTO Manager  
VALUES('M001','Abdulaziz','Faisal',25000.00,'Dept001',0573563567,'Abdulazizfaisal@h  
ospital.com' );
```

```
INSERT INTO Manager  
VALUES('M002','Maria','Ali',22000.00,'Dept002',0563643994,'Mariaali@hospital.com' );
```

```
INSERT INTO Manager  
VALUES('M003','Saad','Sami',28000.00,'Dept003',0589348426,'Saadsami@hospital.com' );
```

```
INSERT INTO Manager  
VALUES('M004','Aseel','Ahmed',26000.00,'Dept004',0588768426,'Aseelahmed@hospit al.com' );
```

```
INSERT INTO Manager  
VALUES('M005','Maher','Salem',27000.00,'Dept005',0533898694,'Mahersalem@hospit al.com' );
```

MANAGERID	FIRSTNAME	LASTNAME	SALARY	DEPTNUM	PHONENUMBER	EMAIL
M001	Abdulaziz	Faisal	25000	Dept001	575563567	Abdulazizfaisal@hospital.com
M002	Maria	Ali	22000	Dept002	563643994	Mariaali@hospital.com
M003	Saad	Sami	28000	Dept003	589348426	Saadsami@hospital.com
M005	Maher	Salem	27000	Dept005	533898694	Mahersalem@hospital.com
M004	Aseel	Ahmed	26000	Dept004	589788426	Aseelahmed@hospital.com



Data Definition Language(DDL)

```
CREATE TABLE Doctor (  
  doctorID CHAR(6),  
  firstName VARCHAR(25) NOT NULL ,  
  lastName VARCHAR(50) NOT NULL , Specialization  
  VARCHAR(50) NOT NULL ,  
  Salary DECIMAL(8,2) CHECK(Salary >= 7000) ,  
  deptNum CHAR(7),  
  phoneNumber INT UNIQUE NOT NULL ,  
  Email VARCHAR (50) ,  
  CONSTRAINT pkDoctor PRIMARY KEY (doctorID),  
  CONSTRAINT fkDeptDoc FOREIGN KEY (deptNum)  
  REFERENCES Department (deptNum) );
```

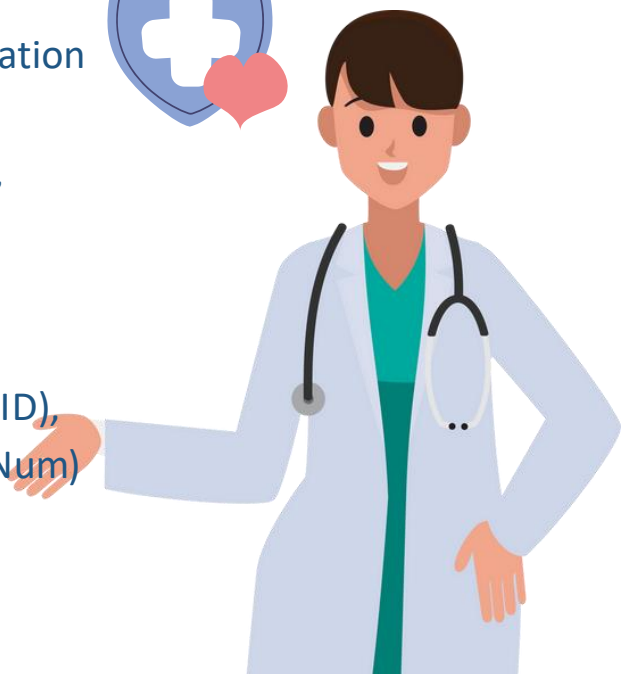


Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DOCTOR	<u>DOCTORID</u>	CHAR	6	-	-	1	-	-	-
	<u>FIRSTNAME</u>	VARCHAR2	25	-	-	-	-	-	-
	<u>LASTNAME</u>	VARCHAR2	50	-	-	-	-	-	-
	<u>SPECIALIZATION</u>	VARCHAR2	50	-	-	-	-	-	-
	<u>SALARY</u>	NUMBER	-	8	2	-	✓	-	-
	<u>DEPTNUM</u>	CHAR	7	-	-	-	✓	-	-
	<u>PHONENUMBER</u>	NUMBER	22	-	0	-	-	-	-
	<u>EMAIL</u>	VARCHAR2	50	-	-	-	✓	-	-
1 - 8									



Data Definition Language(DDL)

```
INSERT INTO Doctor
VALUES('D001' , 'Yusuf' , 'Khaled' , 'Electrophysiologist' , 8500.00 , 'Dept004' , 0554567890 , NULL);
INSERT INTO Doctor
VALUES('D002' , 'Omar' , 'Hassan' , 'Spinal Cord Injury Specialist' , 9200.00 , 'Dept002' , 0505678901 , 'ohassan@hospital.co m');
INSERT INTO Doctor
VALUES('D003' , 'Aisha' , 'Mohammed' , 'Addiction Psychiatrist' , 7800.00 , 'Dept003' , 0556789012 , 'amohammed@hospital.com');
INSERT INTO Doctor
VALUES('D004' , 'Ali' , 'Ibrahim' , 'Cardiothoracic Surgeon' , 10500.00 , 'Dept004' , 0557890123 , 'aibrahim@hospital.com'); INSERT
INTO Doctor
VALUES('D005' , 'Fatima' , 'Abdullah' , 'Oral and Maxillofacial Surgeon' , 8000.00 , 'Dept005' , 0558901234 ,
'fabdullah@hospital.com');
INSERT INTO Doctor
VALUES('D006' , 'Sarah' , 'Ahmed' , 'Ophthalmic Pathologist' , 8200.00 , 'Dept001' , 0509012345 , NULL);
INSERT INTO Doctor
VALUES('D007' , 'Khalid' , 'Ali' , 'Spinal Cord Injury Specialist' , 9000.00 , 'Dept002' , 0500123456 , 'khalia@hospital.com'); INSERT
INTO Doctor
VALUES('D008' , 'Miriam' , 'Omar' , 'Ophthalmic Pathologist' , 7500.00 , 'Dept001' , 0551234567 , 'momar@hospital.com'); INSE RT
INTO Doctor
VALUES('D009' , 'Ahmed' , 'saud' , 'Cardiothoracic Surgeon' , 10200.00 , 'Dept004' , 0502345678 ,
'afatima@hospital.com');
INSERT INTO Doctor
VALUES('D010' , 'Noora' , 'Khaled' , 'Orthodontist' , 7800.00 , 'Dept005' , 0553456789 , 'nkhaled@hospital.com');
```

DOCTORID	FIRSTNAME	LASTNAME	SPECIALIZATION	SALARY	DEPTNUM	PHONENUMBER	EMAIL
D001	Yusuf	Khaled	Electrophysiologist	8500	Dept004	554567890	-
D002	Omar	Hassan	Spinal Cord Injury Specialist	9200	Dept002	505678901	ohassan@hospital.com
D003	Aisha	Mohammed	Addiction Psychiatrist	7800	Dept003	556789012	amohammed@hospital.com
D004	Ali	Ibrahim	Cardiothoracic Surgeon	10500	Dept004	557890123	aibrahim@hospital.com
D005	Fatima	Abdullah	Oral and Maxillofacial Surgeon	8000	Dept005	558901234	fabdullah@hospital.com
D006	Sarah	Ahmed	Ophthalmic Pathologist	8200	Dept001	509012345	-
D007	Khalid	Ali	Spinal Cord Injury Specialist	9000	Dept002	500123456	khalia@hospital.com
D008	Miriam	Omar	Ophthalmic Pathologist	7500	Dept001	551234567	momar@hospital.com
D009	Ahmed	saud	Cardiothoracic Surgeon	10200	Dept004	502345678	afatima@hospital.com
D010	Noora	Khaled	Orthodontist	7800	Dept005	553456789	nkhaled@hospital.com



Data Definition Language(DDL)

```
CREATE TABLE Nurses(  
  nurseID CHAR(6),  
  firstName VARCHAR(25) NOT NULL ,  
  lastName VARCHAR(50) NOT NULL ,  
  Salary DECIMAL(7,2) NOT NULL CHECK(Salary  
    between 6000 and 15000) ,  
  Specialization VARCHAR(50) NOT NULL , deptNum  
  CHAR(7),  
  Email VARCHAR (50) ,  
  phoneNumber INT UNIQUE NOT NULL , CONSTRAINT  
  pkNurse PRIMARY KEY (nurseID), CONSTRAINT  
  fkDeptNurse FOREIGN KEY (deptNum) REFERENCES  
  Department (deptNum)  
);
```

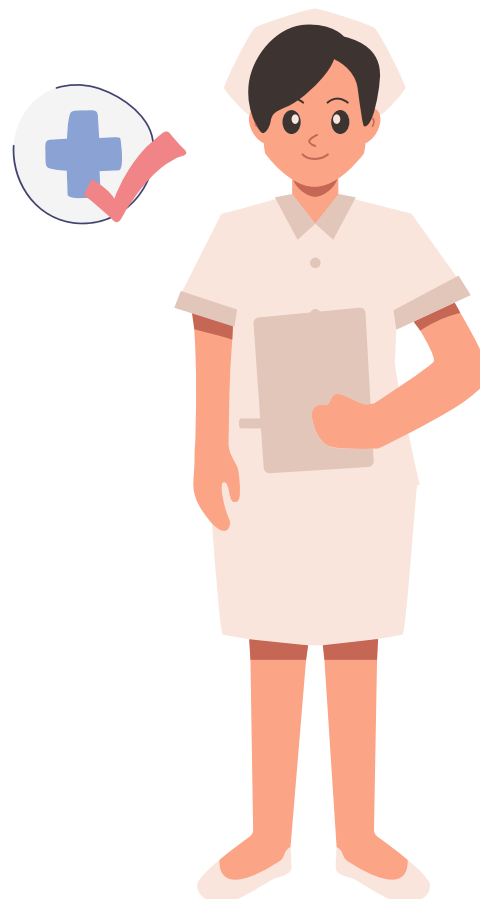


Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>NURSES</u>	<u>NURSEID</u>	CHAR	6	-	-	1	-	-	-
	<u>FIRSTNAME</u>	VARCHAR2	25	-	-	-	-	-	-
	<u>LASTNAME</u>	VARCHAR2	50	-	-	-	-	-	-
	<u>SALARY</u>	NUMBER	-	7	2	-	-	-	-
	<u>SPECIALIZATION</u>	VARCHAR2	50	-	-	-	-	-	-
	<u>DEPTNUM</u>	CHAR	7	-	-	-	✓	-	-
	<u>EMAIL</u>	VARCHAR2	50	-	-	-	✓	-	-
	<u>PHONENUMBER</u>	NUMBER	22	-	0	-	-	-	-
									1 - 8



Data Definition Language(DDL)

```
INSERT INTO Nurses
VALUES('N001', 'Fatima', 'Al-Abdulwahab', 12000.00, 'Ophthalmic Nurse', 'Dept001', 'falwahab@hospital.com',
0554567890);
INSERT INTO Nurses
VALUES('N002', 'Omar', 'Al-Hassan', 11500.00, 'Ophthalmic Nurse', 'Dept001', 'oalhassan@hospital.com', 0555678901);
INSERT INTO Nurses
VALUES('N003', 'Aisha', 'Mohammed', 10000.00, 'Psychiatric Nurse', 'Dept003', 'amohammed@hospital.com',
0506789012);
INSERT INTO Nurses
VALUES('N004', 'Ali', 'Ibrahim', 9500.00, 'Oral surgery nurse', 'Dept005', 'aibrahim@hospital.com', 0507890123); INSERT
INTO Nurses
VALUES('N005', 'Sarah', 'Ahmed', 10000.00, 'Ocular Surgery Nurse', 'Dept001', 'sahmed@hospital.com', 0558901234);
INSERT INTO Nurses
VALUES('N006', 'Khalid', 'Ali', 11000.00, 'Orthopedic Nurse', 'Dept002', 'khalia@hospital.com', 0509012345); INSERT IN TO
Nurses
VALUES('N007', 'Miriam', 'Omar', 9800.00, 'Psychiatric Nurse', 'Dept003', NULL, 0551234567);
INSERT INTO Nurses
VALUES('N008', 'Ahmed', 'saud', 10200.00, 'Cardiac Nurse', 'Dept004', NULL, 0552345678);
INSERT INTO Nurses
VALUES('N009', 'Noora', 'Khaled', 9500.00, 'Orthopedic Nurse', 'Dept002', NULL, 0503456789);
INSERT INTO Nurses
VALUES('N010', 'Laila', 'Mohammed', 10500.00, 'Cardiac Nurse', 'Dept004', NULL, 0502223333);
```

NURSEID	FIRSTNAME	LASTNAME	SALARY	SPECIALIZATION	DEPTNUM	EMAIL	PHONENUMBER
N001	Fatima	Al-Abdulwahab	12000	Ophthalmic Nurse	Dept001	falwahab@hospital.com	554567890
N002	Omar	Al-Hassan	11500	Ophthalmic Nurse	Dept001	oalhassan@hospital.com	555678901
N003	Aisha	Mohammed	10000	Psychiatric Nurse	Dept003	amohammed@hospital.com	506789012
N004	Ali	Ibrahim	9500	Oral surgery nurse	Dept005	aibrahim@hospital.com	507890123
N005	Sarah	Ahmed	10000	Ocular Surgery Nurse	Dept001	sahmed@hospital.com	558901234
N006	Khalid	Ali	11000	Orthopedic Nurse	Dept002	khalia@hospital.com	509012345
N007	Miriam	Omar	9800	Psychiatric Nurse	Dept003	-	551234567
N008	Ahmed	saud	10200	Cardiac Nurse	Dept004	-	552345678
N009	Noora	Khaled	9500	Orthopedic Nurse	Dept002	-	503456789
N010	Laila	Mohammed	10500	Cardiac Nurse	Dept004	-	502223333



Data Definition Language(DDL)

```
CREATE TABLE Room (  
  roomNum CHAR(8),  
  Ward VARCHAR(50) NOT NULL,  
  Capacity INT CHECK (Capacity >= 2),  
  CONSTRAINT pkRoomNum PRIMARY  
  KEY (roomNum)  
);
```



Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ROOM	ROOMNUM	CHAR	8	-	-	1	-	-	-
	WARD	VARCHAR2	50	-	-	-	-	-	-
	CAPACITY	NUMBER	22	-	0	-	✓	-	-
1 - 3									



Data Definition Language(DDL)

```
INSERT INTO Room
VALUES ('Room001', 'ICU', 5);
INSERT INTO Room
VALUES ('Room002', 'Orthopedics', 4);
INSERT INTO Room
VALUES ('Room003', 'Dialysis', 6); INSERT
INTO Room
VALUES ('Room004', 'Cardiology', 3);
INSERT INTO Room
VALUES ('Room005', 'General', 7); INSERT
INTO Room
VALUES ('Room006', 'Burns', 4); INSERT
INTO Room
VALUES ('Room007', 'General', 6); INSERT
INTO Room
VALUES ('Room008', 'Orthopedics', 5);
INSERT INTO Room
VALUES ('Room009', 'ICU', 4);
INSERT INTO Room
VALUES ('Room010', 'Cardiology', 4);
```



ROOMNUM	WARD	CAPACITY
Room001	ICU	5
Room002	Orthopedics	4
Room003	Dialysis	6
Room004	Cardiology	3
Room005	General	7
Room006	Burns	4
Room007	General	6
Room008	Orthopedics	5
Room009	ICU	4
Room010	Cardiology	4



Data Definition Language(DDL)

```
CREATE TABLE Employee (  
  employeeID CHAR(7),  
  firstName VARCHAR(25) NOT NULL,  
  lastName VARCHAR(50) NOT NULL,  
  Position VARCHAR(50) NOT NULL,  
  Salary DECIMAL(7,2) NOT NULL CHECK (Salary  
  BETWEEN 6000 AND 30000),  
  deptNum CHAR(7),  
  phoneNumber INT UNIQUE NOT NULL,  
  CONSTRAINT pkEmpID PRIMARY KEY  
  (employeeID),  
  CONSTRAINT fkDeptEmp FOREIGN KEY  
  (deptNum) REFERENCES  
  Department(deptNum)  
);
```



Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEE	EMPLOYEEID	CHAR	7	-	-	1	-	-	-
	FIRSTNAME	VARCHAR2	25	-	-	-	-	-	-
	LASTNAME	VARCHAR2	50	-	-	-	-	-	-
	POSITION	VARCHAR2	50	-	-	-	-	-	-
	SALARY	NUMBER	-	7	2	-	-	-	-
	DEPTNUM	CHAR	7	-	-	-	✓	-	-
	PHONENUMBER	NUMBER	22	-	0	-	-	-	-
1 - 7									



Data Definition Language(DDL)

```
INSERT INTO Employee
VALUES ('Emp001', 'Ahmed', 'Faris', 'Receptionist', 8000, 'Dept001', 0598765432); INSERT
INTO Employee
VALUES ('Emp002', 'Fatima', 'Ali', 'Administrator', 10000, 'Dept002', 0509876543); INSERT
INTO Employee
VALUES ('Emp003', 'Nawaf', 'Bader', 'Finance Analyst', 9000, 'Dept004', 0512345678); INSERT
INTO Employee
VALUES ('Emp004', 'Sara', 'Ahmed', 'IT Specialist', 13000, 'Dept004', 0501234567); INSERT
INTO Employee
VALUES ('Emp005', 'Omar', 'Yousef', 'Security Officer', 7000, 'Dept005', 0523456789); INSERT
INTO Employee
VALUES ('Emp006', 'LayIn', 'Faisal', 'Receptionist', 8000, 'Dept002', 0534567890); INSERT INTO
Employee
VALUES ('Emp007', 'Khaled', 'Fahad', 'Finance Analyst', 12000, 'Dept003', 0545678901);
INSERT INTO Employee
VALUES ('Emp008', 'Lana', 'Saleh', 'HR Manager', 11000, 'Dept001', 0505273446); INSERT INTO
Employee
VALUES ('Emp009', 'Naser', 'Ali', 'HR Manager', 11000, 'Dept002', 0593767740);
INSERT INTO Employee
VALUES ('Emp010', 'Nora', 'Majed', 'Finance Analyst', 12000, 'Dept003', 0541157177);
```

EMPLOYEEID	FIRSTNAME	LASTNAME	POSITION	SALARY	DEPTNUM	PHONENUMBER
Emp001	Ahmed	Faris	Receptionist	8000	Dept001	598765432
Emp002	Fatima	Ali	Administrator	10000	Dept002	509876543
Emp003	Nawaf	Bader	Finance Analyst	9000	Dept004	512345678
Emp004	Sara	Ahmed	IT Specialist	13000	Dept004	501234567
Emp005	Omar	Yousef	Security Officer	7000	Dept005	523456789
Emp006	LayIn	Faisal	Receptionist	8000	Dept002	534567890
Emp007	Khaled	Fahad	Finance Analyst	12000	Dept003	545678901
Emp008	Lana	Saleh	HR Manager	11000	Dept001	505273446
Emp009	Naser	Ali	HR Manager	11000	Dept002	593767740
Emp010	Nora	Majed	Finance Analyst	12000	Dept003	541157177



Data Definition Language(DDL)

```
CREATE TABLE Appointment (  
  appointmentID CHAR(6),  
  patientID CHAR(10),  
  doctorID CHAR(6),  
  deptNum CHAR(7),  
  appointmentDate TIMESTAMP(0) NOT NULL,  
  CONSTRAINT pkAppoint PRIMARY  
  KEY(appointmentID),  
  CONSTRAINT fkAppointPatient FOREIGN KEY  
  (patientID) REFERENCES Patient(patientID),  
  CONSTRAINT fkAppointDoctor FOREIGN KEY  
  (doctorID) REFERENCES Doctor(doctorID),  
  CONSTRAINT fkAppointmentDept FOREIGN KEY  
  (deptNum) REFERENCES Department(deptNum)  
);
```



Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
APPOINTMENT	APPOINTMENTID	CHAR	6	-	-	1	-	-	-
	PATIENTID	CHAR	10	-	-	-	✓	-	-
	DOCTORID	CHAR	6	-	-	-	✓	-	-
	DEPTNUM	CHAR	7	-	-	-	✓	-	-
	APPOINTMENTDATE	TIMESTAMP(0)	7	-	0	-	-	-	-
									1 - 5



Data Definition Language(DDL)

INSERT INTO Appointment

VALUES ('A001', 'P001', 'D001', 'Dept001',TIMESTAMP '2024-03-10 10:00:00'); INSERT
INTO Appointment

VALUES ('A002', 'P002', 'D002', 'Dept002',TIMESTAMP '2024-03-11 11:00:00'); INSERT
INTO Appointment

VALUES ('A003', 'P003', 'D003', 'Dept003',TIMESTAMP '2024-03-12 12:00:00'); INSERT
INTO Appointment

VALUES ('A004', 'P004', 'D004', 'Dept004',TIMESTAMP '2024-03-13 13:00:00'); INSERT
INTO Appointment

VALUES ('A005', 'P005', 'D005', 'Dept005',TIMESTAMP '2024-03-14 14:00:00'); INSERT
INTO Appointment

VALUES ('A006', 'P006', 'D006', 'Dept001',TIMESTAMP '2024-03-15 15:00:00');

INSERT INTO Appointment

VALUES ('A007', 'P007', 'D007', 'Dept002',TIMESTAMP '2024-03-16 16:00:00');

APPOINTMENTID	PATIENTID	DOCTORID	DEPTNUM	APPOINTMENTDATE
A001	P001	D001	Dept001	10-MAR-24 10.00.00 AM
A002	P002	D002	Dept002	11-MAR-24 11.00.00 AM
A003	P003	D003	Dept003	12-MAR-24 12.00.00 PM
A004	P004	D004	Dept004	13-MAR-24 01.00.00 PM
A005	P005	D005	Dept005	14-MAR-24 02.00.00 PM
A006	P006	D006	Dept001	15-MAR-24 03.00.00 PM
A007	P007	D007	Dept002	16-MAR-24 04.00.00 PM



Data Definition Language(DDL)

```
CREATE TABLE Admission(  
  admissionID CHAR (7),  
  patientID CHAR (10),  
  roomNum CHAR(8),  
  admissionDate TIMESTAMP(0) NOT NULL,  
  dischargeDate TIMESTAMP(0),  
  CONSTRAINT pkAdmission PRIMARY KEY  
  (admissionID),  
  CONSTRAINT fkAdmissionPatient FOREIGN KEY  
  (patientID) REFERENCES Patient (patientID),  
  CONSTRAINT fkAdmissionRoom FOREIGN KEY  
  (roomNum) REFERENCES Room (roomNum)  
);
```



Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ADMISSION	ADMISSIONID	CHAR	7	-	-	1	-	-	-
	PATIENTID	CHAR	10	-	-	-	✓	-	-
	ROOMNUM	CHAR	8	-	-	-	✓	-	-
	ADMISSIONDATE	TIMESTAMP(0)	7	-	0	-	-	-	-
	DISCHARGEDATE	TIMESTAMP(0)	7	-	0	-	✓	-	-
1 - 5									



Data Definition Language(DDL)

```
INSERT INTO Admission
VALUES ('ADM001','P008','Room001', TIMESTAMP '2023-12-19 09:00:04 ', TIMESTAMP '2023-12-20 09:45:19');
INSERT INTO Admission
VALUES ('ADM002','P009','Room002', TIMESTAMP '2023-12-19 09:30:20', TIMESTAMP '2023-12-19 10:30:11');
INSERT INTO Admission
VALUES ('ADM003','P010','Room003', TIMESTAMP '2023-12-25 11:12:00', TIMESTAMP '2023-12-27 12:10:09');
INSERT INTO Admission
VALUES ('ADM004','P011','Room004', TIMESTAMP '2024-1-5 13:00:14', TIMESTAMP '2024-1-7 13:20:10'); INSERT
INTO Admission
VALUES ('ADM005','P012', 'Room005',TIMESTAMP '2024-1-7 08:10:00',null);
INSERT INTO Admission
VALUES ('ADM006','P013','Room006', TIMESTAMP '2024-1-18 08:30:00 ', TIMESTAMP '2024-2-15
11:20:00');
INSERT INTO Admission
VALUES ('ADM007','P014','Room007', TIMESTAMP '2024-2-1 12:40:00',null);
INSERT INTO Admission
VALUES ('ADM008','P015','Room008', TIMESTAMP '2024-2-20 10:12:00', TIMESTAMP '2024-2-23 11:15:00');
```

ADMISSIONID	PATIENTID	ROOMNUM	ADMISSIONDATE	DISCHARGEDATE
ADM001	P008	Room001	19-DEC-23 09.00.04 AM	20-DEC-23 09.45.19 AM
ADM002	P009	Room002	19-DEC-23 09.30.20 AM	19-DEC-23 10.30.11 AM
ADM003	P010	Room003	25-DEC-23 11.12.00 AM	27-DEC-23 12.10.09 PM
ADM004	P011	Room004	05-JAN-24 01.00.14 PM	07-JAN-24 01.20.10 PM
ADM005	P012	Room005	07-JAN-24 08.10.00 AM	-
ADM006	P013	Room006	18-JAN-24 08.30.00 AM	15-FEB-24 11.20.00 AM
ADM007	P014	Room007	01-FEB-24 12.40.00 PM	-
ADM008	P015	Room008	20-FEB-24 10.12.00 AM	23-FEB-24 11.15.00 AM



Data Definition Language(DDL)

```
CREATE TABLE Prescription(  
PrescriptionID CHAR(7) ,  
PatientID CHAR(10) ,  
doctorID CHAR(6),  
prescriptionDate TIMESTAMP(0) NOT NULL,  
Diagnosis VARCHAR (225) NOT NULL, Medication  
VARCHAR(225) NOT NULL, CONSTRAINT  
pkPrescrep PRIMARY KEY (prescriptionID),  
CONSTRAINT fkPrescriptionPatient FOREIGN KEY  
(patientID) REFERENCES patient(patientID),  
CONSTRAINT fkPrescriptionDoctor FOREIGN KEY  
(doctorID) REFERENCES Doctor(doctorID) );
```



Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PREScription	PREScriptionID	CHAR	7	-	-	1	-	-	-
	PATIENTID	CHAR	10	-	-	-	✓	-	-
	DOCTORID	CHAR	6	-	-	-	✓	-	-
	PREScriptionDATE	TIMESTAMP(0)	7	-	0	-	-	-	-
	DIAGNOSIS	VARCHAR2	225	-	-	-	-	-	-
	MEDICATION	VARCHAR2	225	-	-	-	-	-	-
1 - 6									



Data Definition Language(DDL)

```
INSERT INTO Prescription
VALUES ('PR001', 'P001', 'D006',TIMESTAMP '2024-03-10 09:45:00 ', 'Ophthalmic issue', 'Eye drops'); INSERT
INTO Prescription
VALUES ('PR002', 'P002', 'D002',TIMESTAMP '2024-03-11 15:30:00', 'Spinal cord injury', 'Pain medication');
INSERT INTO Prescription
VALUES ('PR003', 'P003', 'D003', TIMESTAMP '2024-03-12 14:00:00', 'Addiction', 'Behavioral therapy');
INSERT INTO Prescription
VALUES ('PR004', 'P004', 'D004', TIMESTAMP '2024-03-13 10:15:00', 'Heart condition', 'Medication for
arrhythmia');
INSERT INTO Prescription
VALUES ('PR005', 'P005', 'D005', TIMESTAMP '2024-03-14 05:40:00', 'Dental issue', 'Tooth extraction');
INSERT INTO Prescription
VALUES ('PR006', 'P006', 'D006', TIMESTAMP '2024-03-15 07:20:00', 'Ophthalmic issue', 'Eye ointment');
INSERT INTO Prescription
VALUES ('PR007', 'P007', 'D007', TIMESTAMP '2024-03-16 11:00:00', 'Orthopedic issue', 'Ankle brace');
```

PRESCRIPTIONID	PATIENTID	DOCTORID	PRESCRIPTIONDATE	DIAGNOSIS	MEDICATION
PR001	P001	D006	10-MAR-24 09.45.00 AM	Ophthalmic issue	Eye drops
PR002	P002	D002	11-MAR-24 03.30.00 PM	Spinal cord injury	Pain medication
PR003	P003	D003	12-MAR-24 02.00.00 PM	Addiction	Behavioral therapy
PR004	P004	D004	13-MAR-24 10.15.00 AM	Heart condition	Medication for arrhythmia
PR005	P005	D005	14-MAR-24 05.40.00 AM	Dental issue	Tooth extraction
PR006	P006	D006	15-MAR-24 07.20.00 AM	Ophthalmic issue	Eye ointment
PR007	P007	D007	16-MAR-24 11.00.00 AM	Orthopedic issue	Ankle brace



Data Definition Language(DDL)

View 1

```
CREATE VIEW PatientBasicInfo AS
```

```
SELECT patientID, firstName, lastName, Age, Gender  
FROM Patient
```

```
WHERE Age >= 30
```

```
ORDER BY Age DESC
```

```
WITH CHECK OPTION;
```

Output :

PATIENTID	FIRSTNAME	LASTNAME	AGE	GENDER
P005	Abdulaziz	Ali	50	M
P010	Khaled	Ali	48	M
P003	Mohammed	Ahmed	42	M
P013	Nour	Ahmed	39	F
P008	Abdulrahman	Ahmed	38	M
P015	Lama	Ali	36	F
P001	Abdullah	Mohammed	35	M
P012	Abdullah	Ali	33	M
P006	Noura	Khaled	31	F



Data Definition Language(DDL)

View 2

```
CREATE VIEW View_Employee_Department AS  
SELECT e.employeeID, e.firstName, e.lastName, e.Position,  
d.deptNum, d.deptName, d.location  
FROM Employee e  
JOIN Department d ON e.deptNum = d.deptNum  
WHERE d.deptNum IN ('Dept001', 'Dept002', 'Dept003');
```

Output :

EMPLOYEEID	FIRSTNAME	LASTNAME	POSITION	DEPTNUM	DEPTNAME	LOCATION
Emp001	Ahmed	Faris	Receptionist	Dept001	Ophthalmology Department	Floor 2
Emp002	Fatima	Ali	Administrator	Dept002	Orthopedics Department	Floor 2
Emp006	LayIn	Faisal	Receptionist	Dept002	Orthopedics Department	Floor 2
Emp007	Khaled	Fahad	Finance Analyst	Dept003	Psychiatry Department	Floor 1
Emp008	Lana	Saleh	HR Manager	Dept001	Ophthalmology Department	Floor 2
Emp009	Naser	Ali	HR Manager	Dept002	Orthopedics Department	Floor 2
Emp010	Nora	Majed	Finance Analyst	Dept003	Psychiatry Department	Floor 1



Data Definition Language(DDL)

View 3

Create view doctorimportantinfo AS select
firstName,lastName,specialization from
doctor

where salary >8000.00

order by firstName DESC ;

Output:

FIRSTNAME	LASTNAME	SPECIALIZATION
Yusuf	Khaled	Electrophysiologist
Sarah	Ahmed	Ophthalmic Pathologist
Omar	Hassan	Spinal Cord Injury Specialist
Khalid	Ali	Spinal Cord Injury Specialist
Ali	Ibrahim	Cardiothoracic Surgeon
Ahmed	saud	Cardiothoracic Surgeon



Data Definition Language(DDL)

View 4

```
CREATE VIEW DoctorSpecialization AS  
SELECT doctorID, firstName, lastName, Specialization  
FROM Doctor  
WHERE Specialization LIKE '%Surgeon%';
```

Output:

DOCTORID	FIRSTNAME	LASTNAME	SPECIALIZATION
D004	Ali	Ibrahim	Cardiothoracic Surgeon
D005	Fatima	Abdullah	Oral and Maxillofacial Surgeon
D009	Ahmed	saud	Cardiothoracic Surgeon



Data Definition Language(DDL)

View 5

```
CREATE VIEW DepartmentSalaryTotal AS  
SELECT d.deptNum, d.deptName, SUM(e.Salary) AS  
TotalSalary  
FROM Department d  
JOIN Employee e ON d.deptNum = e.deptNum GROUP  
BY d.deptNum, d.deptName;
```

Output:

DEPTNUM	DEPTNAME	TOTALSALARY
Dept001	Ophthalmology Department	19000
Dept005	Dental Department	7000
Dept002	Orthopedics Department	29000
Dept004	Cardiology Department	22000
Dept003	Psychiatry Department	24000



Data Manipulation Language(DML)

Query 1

- show patient's id, first name, and last name, along with the appointment ID and date of appointments

```
SELECT patient.patientid, patient.firstname, patient.lastname,  
Appointment.appointmentID, Appointment.patientID, Appointment.appointmentDate  
FROM patient
```

```
JOIN Appointment ON patient.patientid = Appointment.patientid;
```

-Output:

PATIENTID	FIRSTNAME	LASTNAME	APPOINTMENTID	PATIENTID	APPOINTMENTDATE
P001	Abdullah	Mohammed	A001	P001	10-MAR-24 10.00.00 AM
P002	Fatima	Ali	A002	P002	11-MAR-24 11.00.00 AM
P003	Mohammed	Ahmed	A003	P003	12-MAR-24 12.00.00 PM
P004	Layla	Salman	A004	P004	13-MAR-24 01.00.00 PM
P005	Abdulaziz	Ali	A005	P005	14-MAR-24 02.00.00 PM
P006	Noura	Khaled	A006	P006	15-MAR-24 03.00.00 PM
P007	Sara	Abdullah	A007	P007	16-MAR-24 04.00.00 PM

Query 2

- display the nurse IDs and salaries of nurses in the second department whose salaries are less than the salary of some nurses in the first department

```
select nurseid,salary
```

```
from nurses
```

```
where deptnum='Dept002' and salary < some(select salary from nurses where  
deptnum='Dept001');
```

-Output:

NURSEID	SALARY
N006	11000
N009	9500



Data Manipulation Language(DML)

Query 3

-Display the first name, last name, and position of employees whose first name starts with the letter 'N'.

```
SELECT firstname, lastname, position  
FROM employee  
WHERE firstname LIKE 'N%';
```

-Output:

FIRSTNAME	LASTNAME	POSITION
Nawaf	Bader	Finance Analyst
Naser	Ali	HR Manager
Nora	Majed	Finance Analyst

Query 4

-update the salary of the managers,if the salary is more than 25000 then add 1000, else add 2000

update manager

set salary = case when salary >=25000 then salary + 1000

else salary + 2000

end;

-Output:Before Update :After Update :

MANAGERID	SALARY
M001	25000
M002	22000
M003	28000
M005	27000
M004	26000

MANAGERID	SALARY
M001	26000
M002	24000
M003	29000
M005	28000
M004	27000



Data Manipulation Language(DML)

Query 5

-display employee's first name,last name,department number and salary where their salary is more than 9000 or they work in the first department
`select
firstname,lastname,salary,deptnum
from employee
where salary > 9000 or deptnum ='Dept001';`
-Output:

FIRSTNAME	LASTNAME	SALARY	DEPTNUM
Ahmed	Faris	8000	Dept001
Fatima	Ali	10000	Dept002
Sara	Ahmed	13000	Dept004
Khaled	Fahad	12000	Dept003
Lana	Saleh	11000	Dept001
Naser	Ali	11000	Dept002
Nora	Majed	12000	Dept003

Query 6

- show the patient ID, first name, last name, room number, and admission date for patients who are admitted to the hospital.
`select patientid,firstname,lastname,roomnum,admissiondate
from patient inner join admission using(patientid);`
-Output:

PATIENTID	FIRSTNAME	LASTNAME	ROOMNUM	ADMISSIONDATE
P008	Abdulrahman	Ahmed	Room001	19-DEC-23 09.00.04 AM
P009	Fatima	Mohammed	Room002	19-DEC-23 09.30.20 AM
P010	Khaled	Ali	Room003	25-DEC-23 11.12.00 AM
P011	Amal	Abdulrahman	Room004	05-JAN-24 01.00.14 PM
P012	Abdullah	Ali	Room005	07-JAN-24 08.10.00 AM
P013	Nour	Ahmed	Room006	18-JAN-24 08.30.00 AM
P014	Mohammed	Abdullah	Room007	01-FEB-24 12.40.00 PM
P015	Lama	Ali	Room008	20-FEB-24 10.12.00 AM



Data Manipulation Language(DML)

Query 7

-Display all specializations from the doctors' table.

```
SELECT ALL Specialization  
FROM doctor
```

-Output:

SPECIALIZATION
Electrophysiologist
Spinal Cord Injury Specialist
Addiction Psychiatrist
Cardiothoracic Surgeon
Oral and Maxillofacial Surgeon
Ophthalmic Pathologist
Spinal Cord Injury Specialist
Ophthalmic Pathologist
Cardiothoracic Surgeon
Orthodontist

Query 8

-Display the Nurse ID and their specialty for all nurses whose names end with 'ed' and whose salary is greater than or equal to 10000.

```
SELECT nurseID , specialization  
FROM Nurses  
WHERE lastName LIKE '%ed' AND salary >= 10000;
```

-Output:

NURSEID	SPECIALIZATION
N005	Ocular Surgery Nurse
N003	Psychiatric Nurse
N010	Cardiac Nurse



Data Manipulation Language(DML)

Query 9

-Display the employee ID and first name
for all employees whose department
number in dept001 , dept005, and whose
first name has at least four characters.

```
SELECT employeeID , firstName  
FROM Employee  
WHERE deptNum IN ('Dept001' ,  
'Dept005') AND firstName LIKE '____%';
```

-Output:

EMPLOYEEID	FIRSTNAME
Emp001	Ahmed
Emp005	Omar
Emp008	Lana

Query 10

-Display the deptnum of the employee with the highest salary in each
department and order them by department.

```
SELECT deptNum, MAX(salary) AS maxSalary
```

```
FROM Employee  
GROUP BY deptNum;
```

-Output:

DEPTNUM	MAXSALARY
Dept005	7000
Dept003	12000
Dept004	13000
Dept002	11000
Dept001	11000



Data Manipulation Language(DML)

Query 11

-Display the patient ID, first name, and last name for patients who are aged 30 or above and are currently in the hospital.

```
SELECT patientID, firstName , lastName  
FROM Patient
```

WHERE Age >= 30 -Output:

```
AND patientID IN (  
SELECT patientID  
FROM Admission  
WHERE dischargeDate IS NULL  
);
```

PATIENTID	FIRSTNAME	LASTNAME
P012	Abdullah	Ali

Query 12

-Display the employee ID, Deptnum, and salary for employees whose salary is higher than all salaries of employees in the ophthalmology department(Dept001). -SELECT

```
employeeID , deptNum , salary  
FROM Employee
```

WHERE salary > ALL (-Output:

```
SELECT salary  
FROM Employee  
WHERE deptNum = 'Dept001'  
);
```

EMPLOYEEID	DEPTNUM	SALARY
Emp004	Dept004	13000
Emp007	Dept003	12000
Emp010	Dept003	12000



Data Manipulation Language(DML)

Query 13

display all the specializations that
are available in the hospital and
remove duplicates **SELECT**
DISTINCT specialization
FROM doctor;
- Output:

SPECIALIZATION
Addiction Psychiatrist
Orthodontist
Electrophysiologist
Ophthalmic Pathologist
Oral and Maxillofacial Surgeon
Spinal Cord Injury Specialist
Cardiothoracic Surgeon

Query 14

- Display the newly added “Email” column of type VARCHAR in the
employee table
ALTER TABLE employee**Before Alter:**
ADD Email VARCHAR (50);
- Output:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEE	EMPLOYEEID	CHAR	7	-	-	1	-	-	-
	FIRSTNAME	VARCHAR2	25	-	-	-	-	-	-
	LASTNAME	VARCHAR2	50	-	-	-	-	-	-
	POSITION	VARCHAR2	50	-	-	-	-	-	-
	SALARY	NUMBER	-	7	2	-	-	-	-
	DEPTNUM	CHAR	7	-	-	-	✓	-	-
	PHONENUMBER	NUMBER	22	-	0	-	-	-	-
									1 - 7

After Alter:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
NURSES	NURSEID	CHAR	6	-	-	1	-	-	-
	FIRSTNAME	VARCHAR2	25	-	-	-	-	-	-
	LASTNAME	VARCHAR2	50	-	-	-	-	-	-
	SALARY	NUMBER	-	7	2	-	-	-	-
	SPECIALIZATION	VARCHAR2	50	-	-	-	-	-	-
	DEPTNUM	CHAR	7	-	-	-	✓	-	-
	EMAIL	VARCHAR2	50	-	-	-	✓	-	-
	PHONENUMBER	NUMBER	22	-	0	-	-	-	-
									1 - 8



Data Manipulation Language(DML)

Query 15

- Display the specializations and salaries of doctors and nurses whose salaries are below 10,000

```
SELECT Specialization , Salary  
FROM Doctor  
WHERE Salary < 10000  
UNION ALL  
SELECT Specialization , Salary  
FROM Nurses  
WHERE Salary < 10000;
```

- Output:

SPECIALIZATION	SALARY
Electrophysiologist	8500
Spinal Cord Injury Specialist	9200
Addiction Psychiatrist	7800
Oral and Maxillofacial Surgeon	8000
Ophthalmic Pathologist	8200
Spinal Cord Injury Specialist	9000
Ophthalmic Pathologist	7500
Orthodontist	7800
Oral surgery nurse	9500
Psychiatric Nurse	9800
Orthopedic Nurse	9500

Query 16

- Display Nurses names that ends with an a after four digits **SELECT** firstName

```
FROM Nurses  
WHERE firstName LIKE '____a';
```

-Output:

FIRSTNAME
Aisha
Noora
Laila



Data Manipulation Language(DML)

Query 17

-Display the total salary of all doctors in each department

```
SELECT deptNum, SUM(salary) AS total_salary  
FROM Doctor  
GROUP BY deptNum;
```

-Output:

DEPTNUM	TOTAL_SALARY
Dept005	15800
Dept003	7800
Dept004	29200
Dept002	18200
Dept001	15700

Query 18

-Display the number of nurses for each specialization in the Nurses table

```
SELECT Specialization , COUNT(*) AS nursesNum  
FROM Nurses  
GROUP BY Specialization;
```

-Output:

SPECIALIZATION	NURSESNUM
Oral surgery nurse	1
Ophthalmic Nurse	2
Ocular Surgery Nurse	1
Psychiatric Nurse	2
Orthopedic Nurse	2
Cardiac Nurse	2



Data Manipulation Language(DML)

Query 19

-Display doctorID,firstName,lastName,Email for doctors whose names do not start with A

```
SELECT doctorID,firstName,lastName,Email
```

```
FROM Doctor
```

```
WHERE firstName NOT LIKE 'A%';
```

-Output:

DOCTORID	FIRSTNAME	LASTNAME	EMAIL
D001	Yusuf	Khaled	-
D002	Omar	Hassan	ohassan@hospital.com
D005	Fatima	Abdullah	fabdullah@hospital.com
D006	Sarah	Ahmed	-
D007	Khalid	Ali	khali@hospital.com
D008	Miriam	Omar	momar@hospital.com
D010	Noora	Khaled	nkhaled@hospital.com

Query 20

-Display employeeID ,firstName ,lastName ,deptNum and Salary of the employees who work in the fourth department and their salary is more than 10,000

```
SELECT employeeID,firstName,lastName,deptNum,Salary
```

```
FROM Employee
```

```
WHERE deptNum='Dept004'
```

```
INTERSECT
```

```
SELECT employeeID,firstName,lastName,deptNum,Salary
```

```
FROM Employee
```

```
WHERE Salary >10000;
```

-Output:

EMPLOYEEID	FIRSTNAME	LASTNAME	DEPTNUM	SALARY
Emp004	Sara	Ahmed	Dept004	13000



Data Manipulation Language(DML)

Query 21

-Display employeeID, firstName, lastName and Salary whose salaries are between 10,000 and 20,000

```
SELECT employeeID, firstName, lastName, Salary  
FROM Employee  
WHERE Salary BETWEEN 10000 AND 20000;
```

-Output:

EMPLOYEEID	FIRSTNAME	LASTNAME	SALARY
Emp002	Fatima	Ali	10000
Emp004	Sara	Ahmed	13000
Emp007	Khaled	Fahad	12000
Emp008	Lana	Saleh	11000
Emp009	Naser	Ali	11000
Emp010	Nora	Majed	12000

Query 22

-Display nurseID, firstName, lastName, Salary, deptNum for the nurses working in the first and second departments and their salaries

```
SELECT nurseID, firstName, lastName, Salary, deptNum  
FROM Nurses  
WHERE deptNum IN ('Dept001', 'Dept002');
```

-Output:

NURSEID	FIRSTNAME	LASTNAME	SALARY	DEPTNUM
N001	Fatima	Al-Abdulwahab	12000	Dept001
N002	Omar	Al-Hassan	11500	Dept001
N005	Sarah	Ahmed	10000	Dept001
N006	Khalid	Ali	11000	Dept002
N009	Noora	Khaled	9500	Dept002



Data Manipulation Language(DML)

Query 23

- Increase the salary of a manager with the ID (M004) by 2000

UPDATE Manager

SET Salary = Salary+2000

WHERE managerID='M004' AND deptNum='Dept004' ;

-Output:

Before Update :

MANAGERID	FIRSTNAME	LASTNAME	SALARY	DEPTNUM	PHONENUMBER	EMAIL
M004	Aseel	Ahmed	26000	Dept004	589788426	Aseelahmed@hospital.com

After Update :

MANAGERID	FIRSTNAME	LASTNAME	SALARY	DEPTNUM	PHONENUMBER	EMAIL
M004	Aseel	Ahmed	28000	Dept004	589788426	Aseelahmed@hospital.com

Query 24

-Display employeeID,firstName, lastName,deptNum,Position for the employees who working in the third department and their position is Finance Analyst

SELECT employeeID,firstName, lastName,deptNum,Position

FROM Employee

WHERE deptNum='Dept003' AND Position= 'Finance Analyst';

- Output :

EMPLOYEEID	FIRSTNAME	LASTNAME	DEPTNUM	POSITION
Emp007	Khaled	Fahad	Dept003	Finance Analyst
Emp010	Nora	Majed	Dept003	Finance Analyst



Data Manipulation Language(DML)

Query 25

-Display nurseID,firstName,lastName,deptNum for all nurses who work in the hospital departments except those who work in the first department

```
SELECT nurseID,firstName,lastName,deptNum  
FROM Nurses  
WHERE NOT deptNum='Dept001';
```

-Output:

NURSEID	FIRSTNAME	LASTNAME	DEPTNUM
N003	Aisha	Mohammed	Dept003
N004	Ali	Ibrahim	Dept005
N006	Khalid	Ali	Dept002
N007	Miriam	Omar	Dept003
N008	Ahmed	saud	Dept004
N009	Noora	Khaled	Dept002
N010	Laila	Mohammed	Dept004

Query 26

-Display the employees from department 'Dept001' but have a salary not greater than \$8000:

```
SELECT employeeID, firstName, deptNum, salary  
FROM Employee  
WHERE deptNum = 'Dept001'
```

MINUS

```
SELECT employeeID, firstName, deptNum, salary  
FROM Employee  
WHERE salary > 8000; -Output:
```

EMPLOYEEID	FIRSTNAME	DEPTNUM	SALARY
Emp001	Ahmed	Dept001	8000



Data Manipulation Language(DML)

Query 29

-Display and Counts the number of doctors in each department with salaries of \$8000 or more, only including departments with more than 2 such doctors: **SELECT**

deptNum, COUNT(*) AS NumDoctors

FROM Doctor

WHERE Salary >= 8000

GROUP BY deptNum

HAVING COUNT(*) > 2;

-Output:

DEPTNUM	NUMDOCTORS
Dept004	3

Query 30

-Display the first name and last name of patients under 25 years old who haven't been admitted to the hospital:

SELECT firstName, lastName

FROM Patient

WHERE Age < 25

AND patientID NOT IN (SELECT patientID FROM Admission);

-Output:

FIRSTNAME	LASTNAME
Layla	Salman



Data Manipulation Language(DML)

Query 31:

-Display patientIDs,first Names and last Names alongside their admissionIDs and room numbers. Ensuring that all patients records are included in the result even if they have not been admitted to the hospital.

```
SELECT patientID, firstName, lastName, admissionID, roomNum  
FROM Admission  
NATURAL RIGHT OUTER JOIN Patient;
```

-Output:

PATIENTID	FIRSTNAME	LASTNAME	ADMISSIONID	ROOMNUM
P001	Abdullah	Mohammed	-	-
P002	Fatima	Ali	-	-
P003	Mohammed	Ahmed	-	-
P004	Layla	Salman	-	-
P005	Abdulaziz	Ali	-	-
P006	Noura	Khaled	-	-
P007	Sara	Abdullah	-	-
P008	Abdulrahman	Ahmed	ADM001	Room001
P009	Fatima	Mohammed	ADM002	Room002
P010	Khaled	Ali	ADM003	Room003
P011	Amal	Abdulrahman	ADM004	Room004
P012	Abdullah	Ali	ADM005	Room005
P013	Nour	Ahmed	ADM006	Room006
P014	Mohammed	Abdullah	ADM007	Room007
P015	Lama	Ali	ADM008	Room008



Data Manipulation Language(DML)

Query 32:

- Remove the phone
number column from the
employee table

alter table Employee Drop
Column phoneNumber;

-Output:

The table before:

EMPLOYEEID	FIRSTNAME	LASTNAME	POSITION	SALARY	DEPTNUM	PHONENUMBER
Emp001	Ahmed	Faris	Receptionist	8000	Dept001	598765432
Emp002	Fatima	Ali	Administrator	10000	Dept002	509876543
Emp003	Nawaf	Bader	Finance Analyst	9000	Dept004	512345678
Emp004	Sara	Ahmed	IT Specialist	13000	Dept004	501234567
Emp005	Omar	Yousef	Security Officer	7000	Dept005	523456789
Emp006	Layln	Faisal	Receptionist	8000	Dept002	534567890
Emp007	Khaled	Fahad	Finance Analyst	12000	Dept003	545678901
Emp008	Lana	Saleh	HR Manager	11000	Dept001	505273446
Emp009	Naser	Ali	HR Manager	11000	Dept002	593767740
Emp010	Nora	Majed	Finance Analyst	12000	Dept003	541157177

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEE	EMPLOYEEID	CHAR	7	-	-	1	-	-	-
	FIRSTNAME	VARCHAR2	25	-	-	-	-	-	-
	LASTNAME	VARCHAR2	50	-	-	-	-	-	-
	POSITION	VARCHAR2	50	-	-	-	-	-	-
	SALARY	NUMBER	-	7	2	-	-	-	-
	DEPTNUM	CHAR	7	-	-	-	✓	-	-
	PHONENUMBER	NUMBER	22	-	0	-	-	-	-
1 - 7									

The table AFTER:

EMPLOYEEID	FIRSTNAME	LASTNAME	POSITION	SALARY	DEPTNUM
Emp002	Fatima	Ali	Administrator	10000	Dept002
Emp003	Nawaf	Bader	Finance Analyst	9000	Dept004
Emp006	Layln	Faisal	Receptionist	8000	Dept002
Emp008	Lana	Saleh	HR Manager	11000	Dept001
Emp001	Ahmed	Faris	Receptionist	8000	Dept001
Emp004	Sara	Ahmed	IT Specialist	13000	Dept004
Emp005	Omar	Yousef	Security Officer	7000	Dept005
Emp007	Khaled	Fahad	Finance Analyst	12000	Dept003
Emp009	Naser	Ali	HR Manager	11000	Dept002
Emp010	Nora	Majed	Finance Analyst	12000	Dept003

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEE	EMPLOYEEID	CHAR	7	-	-	1	-	-	-
	FIRSTNAME	VARCHAR2	25	-	-	-	-	-	-
	LASTNAME	VARCHAR2	50	-	-	-	-	-	-
	POSITION	VARCHAR2	50	-	-	-	-	-	-
	SALARY	NUMBER	-	7	2	-	-	-	-
	DEPTNUM	CHAR	7	-	-	-	✓	-	-
1 - 6									



Data Manipulation Language(DML)

Query 33

-Display all employee data including employeeID, firstName, lastName, Position, salary, deptNum, phone Number, of employees who have a salary of more than 10,000

SELECT *

FROM Employee

Where salary > 10000;

-Output:

EMPLOYEEID	FIRSTNAME	LASTNAME	POSITION	SALARY	DEPTNUM	PHONENUMBER
Emp004	Sara	Ahmed	IT Specialist	13000	Dept004	501234567
Emp007	Khaled	Fahad	Finance Analyst	12000	Dept003	545678901
Emp008	Lana	Saleh	HR Manager	11000	Dept001	505273446
Emp009	Naser	Ali	HR Manager	11000	Dept002	593767740
Emp010	Nora	Majed	Finance Analyst	12000	Dept003	541157177

Query 34

List the patient's firstName in an ascending order and display their lastName and Gender if it was female

SELECT firstName,lastName ,Gender

FROM Patient

where Gender ='F'

Order by FirstName asc ;

-Output:

FIRSTNAME	LASTNAME	GENDER
Amal	Abdulrahman	F
Fatima	Ali	F
Fatima	Mohammed	F
Lama	Ali	F
Layla	Salman	F
Nour	Ahmed	F
Noura	Khaled	F
Sara	Abdullah	F



Data Manipulation Language(DML)

Query 35

-Display firstName,lastName of nurses who have not registered their emails

```
SELECT firstName,lastName  
FROM nurses
```

```
where Email is null ;
```

-Output:

FIRSTNAME	LASTNAME
Miriam	Omar
Ahmed	saud
Noora	Khaled
Laila	Mohammed

Query 36

-Display firstName ,lastName,Email of female patients who have registered their emails

```
SELECT firstName,lastName ,Email  
from patient
```

```
where Gender= 'F' and Email is not null ;
```

-Output:

FIRSTNAME	LASTNAME	EMAIL
Layla	Salman	layla22@hotmail.com
Noura	Khaled	nourakhaled@email.com
Sara	Abdullah	sara_1997@gmail.com
Fatima	Mohammed	fatima333@email.com
Amal	Abdulrahman	aaamal@email.com
Lama	Ali	lamaali@email.com



Data Manipulation Language(DML)

Query 37

-Display the wards and the total number of
Rooms in each ward

```
SELECT ward,COUNT(roomNum) AS  
Total_Rooms  
from Room  
group by ward; -Output:
```

WARD	TOTAL_ROOMS
General	2
ICU	2
Dialysis	1
Cardiology	2
Orthopedics	2
Burns	1

Query 38

-Display patientID , firstName,
lastName,Gender,age where gender is
M or age is less than 25

```
SELECT patientID , firstName,  
lastName,Gender,age  
FROM patient  
WHERE Gender = 'M'  
UNION  
SELECT patientID , firstName,  
lastName,Gender,age  
FROM patient  
WHERE age < 25;
```

-Output:

PATIENTID	FIRSTNAME	LASTNAME	GENDER	AGE
P001	Abdullah	Mohammed	M	35
P003	Mohammed	Ahmed	M	42
P004	Layla	Salman	F	22
P005	Abdulaziz	Ali	M	50
P008	Abdulrahman	Ahmed	M	38
P010	Khaled	Ali	M	48
P012	Abdullah	Ali	M	33
P014	Mohammed	Abdullah	M	26



Data Manipulation Language(DML)

Query 39

-Combines patient and admission information based on shared patientID.

```
SELECT * FROM Patient NATURAL JOIN Admission;
```

-Output:

PATIENTID	FIRSTNAME	LASTNAME	AGE	GENDER	PHONENUMBER	EMAIL	ADMISSIONID	ROOMNUM	ADMISSIONDATE	DISCHARGEDATE
P008	Abdulrahman	Ahmed	38	M	501851122	-	ADM001	Room001	19-DEC-23 09.00.04 AM	20-DEC-23 09.45.19 AM
P009	Fatima	Mohammed	29	F	597773344	fatima333@email.com	ADM002	Room002	19-DEC-23 09.30.20 AM	19-DEC-23 10.30.11 AM
P010	Khaled	Ali	48	M	523325566	khaled_ali@email.com	ADM003	Room003	25-DEC-23 11.12.00 AM	27-DEC-23 12.10.09 PM
P011	Amal	Abdulrahman	25	F	539997788	aaamal@email.com	ADM004	Room004	05-JAN-24 01.00.14 PM	07-JAN-24 01.20.10 PM
P012	Abdullah	Ali	33	M	523219900	abdullah.ali@email.com	ADM005	Room005	07-JAN-24 08.10.00 AM	-
P013	Nour	Ahmed	39	F	510601122	-	ADM006	Room006	18-JAN-24 08.30.00 AM	15-FEB-24 11.20.00 AM
P014	Mohammed	Abdullah	26	M	579783344	-	ADM007	Room007	01-FEB-24 12.40.00 PM	-
P015	Lama	Ali	36	F	586645566	lamaali@email.com	ADM008	Room008	20-FEB-24 10.12.00 AM	23-FEB-24 11.15.00 AM

Query 40

-Show patient's first names and last names and rename the firstName column as First and lastName column as Last

```
SELECT  
firstName AS First, lastName AS Last FROM Patient;
```

 -Output:

FIRST	LAST
Abdullah	Mohammed
Fatima	Ali
Mohammed	Ahmed
Layla	Salman
Abdulaziz	Ali
Noura	Khaled
Sara	Abdullah
Abdulrahman	Ahmed
Fatima	Mohammed
Khaled	Ali
Amal	Abdulrahman
Abdullah	Ali
Nour	Ahmed
Mohammed	Abdullah
Lama	Ali



Data Manipulation Language(DML)

Query 41

- Lists all the patient's information sorted by age in descending order.

```
SELECT * FROM Patient ORDER BY Age DESC;
```

- Output:

PATIENTID	FIRSTNAME	LASTNAME	AGE	GENDER	PHONENUMBER	EMAIL
P005	Abdulaziz	Ali	50	M	563353344	-
P010	Khaled	Ali	48	M	523325566	khaled_ali@email.com
P003	Mohammed	Ahmed	42	M	544349101	m_ahmed@gmail.com
P013	Nour	Ahmed	39	F	510601122	-
P008	Abdulrahman	Ahmed	38	M	501851122	-
P015	Lama	Ali	36	F	586645566	lamaali@email.com
P001	Abdullah	Mohammed	35	M	535551234	abdullah.mohammed@email.com
P012	Abdullah	Ali	33	M	523219900	abdullah.ali@email.com
P006	Noura	Khaled	31	F	513375566	nourakhaled@email.com
P009	Fatima	Mohammed	29	F	597773344	fatima333@email.com
P002	Fatima	Ali	28	F	547895678	-
P007	Sara	Abdullah	27	F	553059900	sara_1997@gmail.com
P014	Mohammed	Abdullah	26	M	579783344	-
P011	Amal	Abdulrahman	25	F	539997788	aaamal@email.com
P004	Layla	Salman	22	F	522121122	layla22@hotmail.com

Query 42

- Counts the total number of doctors in the database , change the name of the column to "allDoctors".

```
SELECT COUNT(*) AS allDoctors FROM Doctor;
```

- Output:

ALLDOCTORS
10



Data Manipulation Language(DML)

Query 43

- delete rows from the Employee table where the Position is 'Receptionist' and the Salary is less than 9000.

```
DELETE FROM Employee WHERE Position = 'Receptionist' AND Salary < 9000;
```

-Output:

Before deletion

EMPLOYEEID	FIRSTNAME	LASTNAME	POSITION	SALARY	DEPTNUM	PHONENUMBER
Emp001	Ahmed	Faris	Receptionist	8000	Dept001	598765432
Emp002	Fatima	Ali	Administrator	10000	Dept002	509876543
Emp003	Nawaf	Bader	Finance Analyst	9000	Dept004	512345678
Emp004	Sara	Ahmed	IT Specialist	13000	Dept004	501234567
Emp005	Omar	Yousef	Security Officer	7000	Dept005	523456789
Emp006	LayIn	Faisal	Receptionist	8000	Dept002	534567890
Emp007	Khaled	Fahad	Finance Analyst	12000	Dept003	545678901
Emp008	Lana	Saleh	HR Manager	11000	Dept001	505273446
Emp009	Naser	Ali	HR Manager	11000	Dept002	593767740
Emp010	Nora	Majed	Finance Analyst	12000	Dept003	541157177

After deletion

EMPLOYEEID	FIRSTNAME	LASTNAME	POSITION	SALARY	DEPTNUM	PHONENUMBER
Emp002	Fatima	Ali	Administrator	10000	Dept002	509876543
Emp003	Nawaf	Bader	Finance Analyst	9000	Dept004	512345678
Emp004	Sara	Ahmed	IT Specialist	13000	Dept004	501234567
Emp005	Omar	Yousef	Security Officer	7000	Dept005	523456789
Emp007	Khaled	Fahad	Finance Analyst	12000	Dept003	545678901
Emp008	Lana	Saleh	HR Manager	11000	Dept001	505273446
Emp009	Naser	Ali	HR Manager	11000	Dept002	593767740
Emp010	Nora	Majed	Finance Analyst	12000	Dept003	541157177



Data Manipulation Language(DML)

Query 44

- Show patient information with their admission records, ensuring that every patient, whether admitted or not, is included along with their respective admission details

```
SELECT * FROM Patient FULL OUTER JOIN Admission ON  
Patient.patientID = Admission.patientID;
```

-Output:

PATIENTID	FIRSTNAME	LASTNAME	AGE	GENDER	PHONENUMBER	EMAIL	ADMISSIONID	PATIENTID	ROOMNUM	ADMISSIONDATE	DISCHARGEDATE
P001	Abdullah	Mohammed	35	M	535551234	abdullah.mohammed@email.com	-	-	-	-	-
P002	Fatima	Ali	28	F	547895678	-	-	-	-	-	-
P003	Mohammed	Ahmed	42	M	544349101	m_ahmed@gmail.com	-	-	-	-	-
P004	Layla	Salman	22	F	522121122	layla22@hotmail.com	-	-	-	-	-
P005	Abdulaziz	Ali	50	M	563353344	-	-	-	-	-	-
P006	Noura	Khaled	31	F	513375566	nourakhaled@email.com	-	-	-	-	-
P007	Sara	Abdullah	27	F	553059900	sara_1997@gmail.com	-	-	-	-	-
P008	Abdulrahman	Ahmed	38	M	501851122	-	ADM001	P008	Room001	19-DEC-23 09.00.04 AM	20-DEC-23 09.45.19 AM
P009	Fatima	Mohammed	29	F	597773344	fatima333@email.com	ADM002	P009	Room002	19-DEC-23 09.30.20 AM	19-DEC-23 10.30.11 AM
P010	Khaled	Ali	48	M	523325566	khaled_ali@email.com	ADM003	P010	Room003	25-DEC-23 11.12.00 AM	27-DEC-23 12.10.09 PM
P011	Amal	Abdulrahman	25	F	539997788	aaamal@email.com	ADM004	P011	Room004	05-JAN-24 01.00.14 PM	07-JAN-24 01.20.10 PM
P012	Abdullah	Ali	33	M	523219900	abdullah.ali@email.com	ADM005	P012	Room005	07-JAN-24 08.10.00 AM	-
P013	Nour	Ahmed	39	F	510601122	-	ADM006	P013	Room006	18-JAN-24 08.30.00 AM	15-FEB-24 11.20.00 AM
P014	Mohammed	Abdullah	26	M	579783344	-	ADM007	P014	Room007	01-FEB-24 12.40.00 PM	-
P015	Lama	Ali	36	F	586645566	lamaali@email.com	ADM008	P015	Room008	20-FEB-24 10.12.00 AM	23-FEB-24 11.15.00 AM

Query 45

display the first and last name and the specialization of the doctors who have the letter h in their first name

```
select firstName ,lastName,specialization
```

```
from doctor
```

```
where firstName like '%h%';
```

-Output:

FIRSTNAME	LASTNAME	SPECIALIZATION
Aisha	Mohammed	Addiction Psychiatrist
Sarah	Ahmed	Ophthalmic Pathologist
Khalid	Ali	Spinal Cord Injury Specialist
Ahmed	saud	Cardiothoracic Surgeon



Data Manipulation Language(DML)

Query 46

-Display nurseIDs,their first and last names and their salary if their specialization is cardiac nurse and also their salaries are in the range between 10000.00 and 11000.00

```
select nurseID,firstName,lastName ,salary  
from nurses
```

```
where salary between 10000.00 and 11000.00 and specialization  
='Cardiac Nurse';
```

-Output:

NURSEID	FIRSTNAME	LASTNAME	SALARY
N008	Ahmed	saud	10200
N010	Laila	Mohammed	10500

Query 47

-Display nurseIDs and specializations of nurses in Department 3 whose salary is matching with some of the salaries of nurses in department 1

```
SELECT nurseID, specialization
```

```
FROM Nurses
```

```
WHERE deptNum = 'Dept003'
```

```
AND salary = SOME (SELECT salary FROM Nurses WHERE deptNum =  
'Dept001');
```

-Output:

NURSEID	SPECIALIZATION
N003	Psychiatric Nurse



Data Manipulation Language(DML)

Query 48

-Get all patient details along with their prescription if they are available

```
SELECT *  
FROM Patient  
NATURAL LEFT OUTER JOIN Prescription;
```

-Output:

PATIENTID	FIRSTNAME	LASTNAME	AGE	GENDER	PHONENUMBER	EMAIL	PRESCRIPTIONID	DOCTORID	PRESCRIPTIONDATE	DIAGNOSIS	MEDICATION
P001	Abdullah	Mohammed	35	M	535551234	abdullah.mohammed@email.com	PR001	D006	10-MAR-24 09.45.00 AM	Ophthalmic issue	Eye drops
P002	Fatima	Ali	28	F	547895678	-	PR002	D002	11-MAR-24 03.30.00 PM	Spinal cord injury	Pain medication
P003	Mohammed	Ahmed	42	M	544349101	m_ahmed@gmail.com	PR003	D003	12-MAR-24 02.00.00 PM	Addiction	Behavioral therapy
P004	Layla	Salman	22	F	522121122	layla22@hotmail.com	PR004	D004	13-MAR-24 10.15.00 AM	Heart condition	Medication for arrhythmia
P005	Abdulaziz	Ali	50	M	563353344	-	PR005	D005	14-MAR-24 05.40.00 AM	Dental issue	Tooth extraction
P006	Noura	Khaled	31	F	513375566	nourakhaled@email.com	PR006	D006	15-MAR-24 07.20.00 AM	Ophthalmic issue	Eye ointment
P007	Sara	Abdullah	27	F	553059900	sara_1997@gmail.com	PR007	D007	16-MAR-24 11.00.00 AM	Orthopedic issue	Ankle brace
P008	Abdulrahman	Ahmed	38	M	501851122	-	-	-	-	-	-
P009	Fatima	Mohammed	29	F	597773344	fatima333@email.com	-	-	-	-	-
P010	Khaled	Ali	48	M	523325566	khaled_ali@email.com	-	-	-	-	-
P011	Amal	Abdulrahman	25	F	539997788	aaamal@email.com	-	-	-	-	-
P012	Abdullah	Ali	33	M	523219900	abdullah.ali@email.com	-	-	-	-	-
P013	Nour	Ahmed	39	F	510601122	-	-	-	-	-	-
P014	Mohammed	Abdullah	26	M	579783344	-	-	-	-	-	-
P015	Lama	Ali	36	F	586645566	lamaali@email.com	-	-	-	-	-



Data Manipulation Language(DML)

Query 49

-Retrieve the specializations of doctors who work in departments where at least one nurse earns a salary greater than \$10,000.00

```
SELECT specialization  
FROM doctor d  
WHERE EXISTS (  
  SELECT nurseID  
  FROM nurses n  
  WHERE d.deptNum = n.deptNum and salary>10000.00  
);
```

-Output:

SPECIALIZATION
Ophthalmic Pathologist
Ophthalmic Pathologist
Spinal Cord Injury Specialist
Spinal Cord Injury Specialist
Cardiothoracic Surgeon
Cardiothoracic Surgeon
Electrophysiologist



Data Manipulation Language(DML)

Query 50

-Retrieve the nurseIDs and specialization of nurses working in department 'Dept001' whose salary is greater than at least one nurse's salary in department 'Dept003'

```
SELECT nurseID, specialization
```

```
FROM Nurses
```

```
WHERE deptNum = 'Dept001'
```

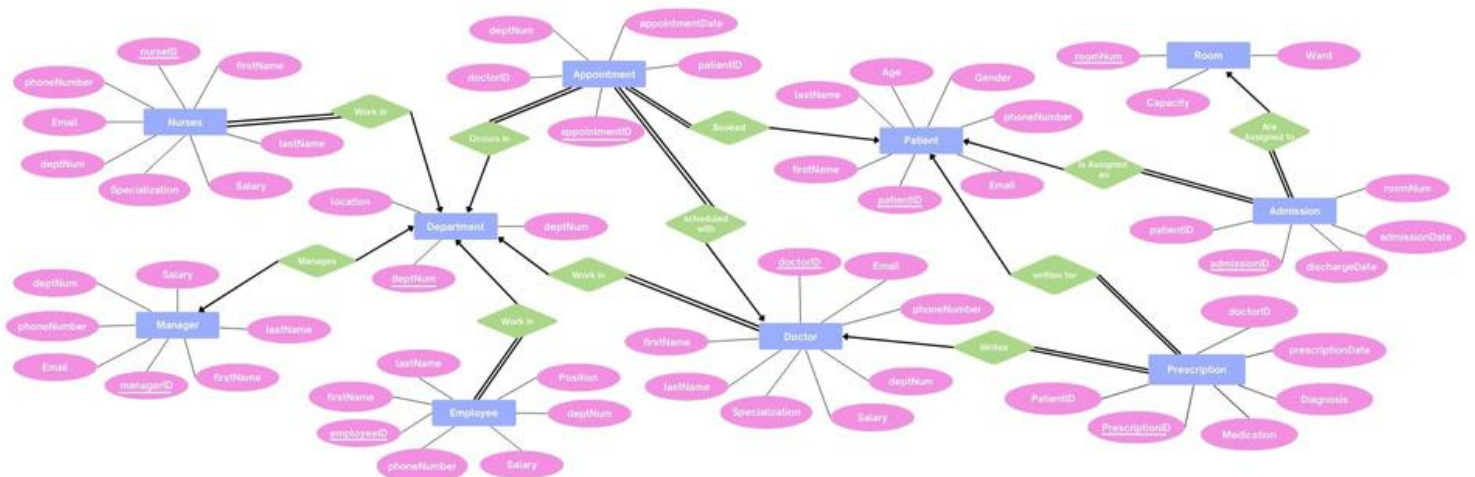
```
AND salary > SOME (SELECT salary FROM Nurses WHERE deptNum =  
'Dept003');
```

-Output :

NURSEID	SPECIALIZATION
N001	Ophthalmic Nurse
N002	Ophthalmic Nurse
N005	Ocular Surgery Nurse



Entity relational Model (ERM)





Normalization Theory



TABLE

ON

Patient_Doctor_Prescription

PATIENTID	PATIENTNAME	PHONENUMBER	DOCTORID	DOCTORNAME	DEPTNUM	DEPTNAME	PRESCRIPTION
P001	Abdullaha Mohammed	535551234	D001	Dr.Yusuf	Dept01	Ophthalmology	Eye Drops
P001	Abdullaha Mohammed	535551234	D002	Dr. Omar	Dept02	Orthopedics	Ankle Brace
P002	Fatima Ali	547895678	D001	Dr.Yusuf	Dept01	Ophthalmology	Eye ointment
P002	Fatima Ali	547895678	D002	Dr. Omar	Dept02	Orthopedics	Pain Medication
P003	Mohammed Ahmed	544349101	D003	Dr.Aisha	Dept03	Psychiatry	Medication for Depression
P004	Layla Salman	522121122	D004	Dr. Ali	Dept04	Cardiology	Medication for Arrhythmia
P005	Abdulaziz Ali	563353344	D005	Dr. Fatima	Dept05	Dental	Pain Relief Pill



Normalization Theory

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PatientInfo

PATIENTID	PATIENTNAME	PHONENUMBER
P001	Abdullah Mohammed	535551234
P002	Fatima Ali	547895678
P003	Mohammed Ahmed	544349101
P004	Layla Salman	522121122
P005	Abdulaziz Ali	563353344

Patient_Prescription

PATIENTID	DOCTORID	DOCTORNAME	DEPTNUM	DEPTNAME	PRESCRIPTION
P001	D001	Dr.Yusuf	Dept01	Ophthalmology	Eye Drops
P001	D002	Dr. Omar	Dept02	Orthopedics	Ankle Brace
P002	D001	Dr.Yusuf	Dept01	Ophthalmology	Eye ointment
P002	D002	Dr. Omar	Dept02	Orthopedics	Pain Medication
P003	D003	Dr.Aisha	Dept03	Psychiatry	Medication for Depression
P004	D004	Dr. Ali	Dept04	Cardiology	Medication for Arrhythmia
P005	D005	Dr. Fatima	Dept05	Dental	Pain Relief Pill



Normalization Theory

2N

PatientInfo

PATIENTID	PATIENTNAME	PHONENUMBER
P001	Abdullah Mohammed	535551234
P002	Fatima Ali	547895678
P003	Mohammed Ahmed	544349101
P004	Layla Salman	522121122
P005	Abdulaziz Ali	563353344

Prescription

PATIENTID	DOCTORID	PRESCRIPTION
P001	D001	Eye Drops
P001	D002	Ankle Brace
P002	D001	Eye ointment
P002	D002	Pain Medication
P003	D003	Medication for Depression
P004	D004	Medication for Arrhythmia
P005	D005	Pain Relief Pill

Doctor_Dept

DOCTORID	DOCTORNAME	DEPTNUM	DEPTNAME
D001	Dr.Yusuf	Dept01	Ophthalmology
D002	Dr. Omar	Dept02	Orthopedics
D003	Dr.Aisha	Dept03	Psychiatry
D004	Dr. Ali	Dept04	Cardiology
D005	Dr. Fatima	Dept05	Dental



Normalization Theory

3N

PatientInfo

PATIENTID	PATIENTNAME	PHONENUMBER
P001	Abdullah Mohammed	535551234
P002	Fatima Ali	547895678
P003	Mohammed Ahmed	544349101
P004	Layla Salman	522121122
P005	Abdulaziz Ali	563353344

DoctorInfo

DOCTORID	DOCTORNAME	DEPTNUM
D001	Dr.Yusuf	Dept01
D002	Dr. Omar	Dept02
D003	Dr.Aisha	Dept03
D004	Dr.Ali	Dept04
D005	Dr.Fatima	Dept05

Prescription

PATIENTID	DOCTORID	PRESCRIPTION
P001	D001	Eye Drops
P001	D002	Ankle Brace
P002	D001	Eye ointment
P002	D002	Pain Medication
P003	D003	Medication for Depression
P004	D004	Medication for Arrhythmia
P005	D005	Pain Relief Pill

Department

DEPTNUM	DEPTNAME
Dept001	Ophthalmology Department
Dept002	Orthopedics Department
Dept003	Psychiatry Department
Dept004	Cardiology Department
Dept005	Dental Department



Normalization Theory



TABLE

ON

Hospital_MedicalProcedures

PATIENTID	PATIENT_NAME	PHONENUMBER	MEDICAL_PROCEDURE_ID	MEDICAL_PROCEDURE_NAME	DEPTNUM	DEPTNAME	BILLS
P001	Abdullah Mohammed	535551234	PR001	Eye examination	Dept001	Ophthalmology Department	150
P002	Fatima Ali	547895678	PR002	Orthopedic consultation	Dept002	Orthopedics Department	200
P003	Mohammed Ahmed	544349101	PR002	Orthopedic consultation	Dept002	Orthopedics Department	190
P003	Mohammed Ahmed	544349101	PR003	Psychiatric evaluation	Dept003	Psychiatry Department	175
P004	Layla Salman	522121122	PR001	Eye examination	Dept001	Ophthalmology Department	130
P004	Layla Salman	522121122	PR004	Cardiac assessment	Dept004	Cardiology Department	250
P005	Abdulaziz Ali	563353344	PR005	Dental examination	Dept005	Dental Department	180



Normalization Theory

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1N

PatientInfo

PATIENTID	PATIENTNAME	PHONENUMBER
P001	Abdullah Mohammed	535551234
P002	Fatima Ali	547895678
P003	Mohammed Ahmed	544349101
P004	Layla Salman	522121122
P005	Abdulaziz Ali	563353344

Patient_MedicalProcedure

PATIENTID	MEDICAL_PROCEDURE_ID	MEDICAL_PROCEDURE_NAME	DEPTNUM	DEPTNAME	BILLS
P001	PR001	Eye examination	Dept001	Ophthalmology Department	150
P002	PR002	Orthopedic consultation	Dept002	Orthopedics Department	200
P003	PR002	Orthopedic consultation	Dept002	Orthopedics Department	190
P003	PR003	Psychiatric evaluation	Dept003	Psychiatry Department	175
P004	PR001	Eye examination	Dept001	Ophthalmology Department	130
P004	PR004	Cardiac assessment	Dept004	Cardiology Department	250
P005	PR005	Dental examination	Dept005	Dental Department	180



Normalization Theory

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PatientInfo

PATIENTID	PATIENTNAME	PHONENUMBER
P001	Abdullaha Mohammed	535551234
P002	Fatima Ali	547895678
P003	Mohammed Ahmed	544349101
P004	Layla Salman	522121122
P005	Abdulaziz Ali	563353344

Patient_Bills

PATIENTID	MEDICAL_PROCEDURE_ID	BILLS
P001	PR001	150
P002	PR002	200
P003	PR002	190
P003	PR003	175
P004	PR001	130
P004	PR004	250
P005	PR005	180

MedicalProcedure_Dept

MEDICAL_PROCEDURE_ID	MEDICAL_PROCEDURE_NAME	DEPTNUM	DEPTNAME
PR001	Eye examination	Dept001	Ophthalmology Department
PR002	Orthopedic consultation	Dept002	Orthopedics Department
PR003	Psychiatric evaluation	Dept003	Psychiatry Department
PR004	Cardiac assessment	Dept004	Cardiology Department
PR005	Dental examination	Dept005	Dental Department



Normalization Theory

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PatientInfo

PATIENTID	PATIENTNAME	PHONENUMBER
P001	Abdullah Mohammed	535551234
P002	Fatima Ali	547895678
P003	Mohammed Ahmed	544349101
P004	Layla Salman	522121122
P005	Abdulaziz Ali	563353344

Patient_Bills

PATIENTID	MEDICAL_PROCEDURE_ID	BILLS
P001	PR001	150
P002	PR002	200
P003	PR002	190
P003	PR003	175
P004	PR001	130
P004	PR004	250
P005	PR005	180

MedicalProcedure_Info

MEDICAL_PROCEDURE_ID	MEDICAL_PROCEDURE_NAME	DEPTNUM
PR001	Eye examination	Dept001
PR002	Orthopedic consultation	Dept002
PR003	Psychiatric evaluation	Dept003
PR004	Cardiac assessment	Dept004
PR005	Dental examination	Dept005

Department

DEPTNUM	DEPTNAME
Dept001	Ophthalmology Department
Dept002	Orthopedics Department
Dept003	Psychiatry Department
Dept004	Cardiology Department
Dept005	Dental Department



Normalization Theory



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Hospital_Patient_Admission

PATIENTID	PATIENTNAME	AGE	ROOMNUM	WARD	NURSEID	NURSENAME	ADMISSIONDATE
P001	Abdullah Mohammed	35	Room001	General	N001	Sarah Ali	12/19/2023
P002	Fatima Ali	28	Room002	Orthopedics	N006	Khalid Ali	12/19/2023
P002	Fatima Ali	28	Room007	General	N001	Sarah Ali	09/05/2024
P003	Mohammed Ahmed	42	Room004	Cardiology	N008	Ahmed Saud	12/25/2023
P003	Mohammed Ahmed	42	Room002	Orthopedics	N006	Khalid Ali	04/19/2024
P004	Layla Salman	22	Room010	Cardiology	N010	Ali Ibrahim	01/05/2024
P005	Abdulaziz Ali	50	Room010	Cardiology	N010	Ali Ibrahim	10/20/2024
P005	Abdulaziz Ali	50	Room008	Orthopedics	N009	Laila Mohammed	01/07/2024



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PatientInfo

PATIENTID	PATIENTNAME	AGE
P001	Abdullah Mohammed	35
P002	Fatima Ali	28
P003	Mohammed Ahmed	42
P004	Layla Salman	22
P005	Abdulaziz Ali	50

AllPatient_Admissions

PATIENTID	ROOMNUM	WARD	NURSEID	NURSENAME	ADMISSIONDATE
P001	Room001	General	N001	Sarah Ali	12/19/2023
P002	Room002	Orthopedics	N006	Khalid Ali	12/19/2023
P002	Room007	General	N001	Sarah Ali	09/05/2024
P003	Room002	Orthopedics	N006	Khalid Ali	04/19/2024
P003	Room004	Cardiology	N008	Ahmed Saud	12/25/2023
P004	Room010	Cardiology	N010	Ali Ibrahim	01/05/2024
P005	Room008	Orthopedics	N009	Laila Mohammed	01/07/2024
P005	Room010	Cardiology	N010	Ali Ibrahim	10/20/2024



Normalization Theory

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PatientInfo

PATIENTID	PATIENTNAME	AGE
P001	Abdullah Mohammed	35
P002	Fatima Ali	28
P003	Mohammed Ahmed	42
P004	Layla Salman	22
P005	Abdulaziz Ali	50

AdmissionDate

PATIENTID	ROOMNUM	ADMISSIONDATE
P001	Room001	12/19/2023
P002	Room002	12/19/2023
P002	Room007	09/05/2024
P003	Room002	04/19/2024
P003	Room004	12/25/2023
P004	Room010	01/05/2024
P005	Room008	01/07/2024
P005	Room010	10/20/2024

Room_Nurses

ROOMNUM	WARD	NURSEID	NURSENAME
Room001	General	N001	Sarah Ali
Room002	Orthopedics	N006	Khalid Ali
Room004	Cardiology	N008	Ahmed Saud
Room007	General	N001	Sarah Ali
Room008	Orthopedics	N009	Laila Mohammed
Room010	Cardiology	N010	Ali Ibrahim



Normalization Theory

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PatientInfo

PATIENTID	PATIENTNAME	AGE
P001	Abdullah Mohammed	35
P002	Fatima Ali	28
P003	Mohammed Ahmed	42
P004	Layla Salman	22
P005	Abdulaziz Ali	50

AdmissionDate

PATIENTID	ROOMNUM	ADMISSIONDATE
P001	Room001	12/19/2023
P002	Room002	12/19/2023
P002	Room007	09/05/2024
P003	Room002	04/19/2024
P003	Room004	12/25/2023
P004	Room010	01/05/2024
P005	Room008	01/07/2024
P005	Room010	10/20/2024

Rooms

ROOMNUM	WARD	NURSEID
Room001	General	N001
Room002	Orthopedics	N006
Room004	Cardiology	N008
Room007	General	N001
Room008	Orthopedics	N009
Room010	Cardiology	N010

Nurses

NURSEID	NURSENAME
N001	Sarah Ali
N006	Khalid Ali
N008	Ahmed Saud
N009	Laila Mohammed
N010	Ali Ibrahim